Engineering Chemistry · Unit_04 · Functional materials and green chemistry

DRAFT — UE26CY151A · Unit_04 · Functional materials and green chemistry · Green chemistry: Introduction, twelve principles of green chemistry with examples

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
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Low ASR confidence	Transcript segment requires audio verification.
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M1

Green chemistry: Introduction, twelve principles of green chemistry with examples

1 Green chemistry: Introduction, twelve principles of green chemistry with examples

Study focus: Green chemistry: Introduction, twelve principles of green chemistry with examples. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000174 · EV0000175

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Today we will look at module eight and that is about green chemistry and this is the only class in this module.

Key details

- We will study the definition and the twelve principles of green chemistry.
- Green chemistry is defined as the utilization of a set of principles that reduces or eliminates the use of use or generation of hazardous substances in the design, manufacture, and application of products.
- We all are aware of environmental chemistry where we talk about removing eliminating pollution removing pollution from the environment.
- But here green chemistry talks about.
- Removing pollution, yes, but not generating it in the first place itself.
- Anistos and John C.
- The what is the focus of green chemistry?
- Then we have to find creative alternative routes to minimize impact on the environment.
- And it also, of course, provides sustainable development, sustainable businesses, and sustainable living practices.
- Prevent waste, atom economy, safe synthesis, safe products, design benign chemicals, design for energy efficiency, use of renewable feed stocks, reduce derivatives, catalysis,.
- Designed for degradation, pollution prevention, and accident prevention.

- Of course the first thing we have to talk about when we talk about pollution is that not to generate waste in the first place because once the waste is produced that it involves an extra cost and time to treat and safely dispose it.
- And therefore the production cost also increases.
- The second principle is atom economy.
- In fact, in such a way the design should be of any reaction should be such that all the materials, in fact all the atoms which are present in the reactants should be incorporated into the products.
- That is atom economy.
- The emphasis was more on yield.
- Yes, here also we want a good yield, but at the same time it is not just the yield we are interested in.
- We want that all the atoms so that there is no extra atoms which are not being used or getting wasted.
- Some of the atoms are getting substituted by another atom.

Example or application from the lecture

- Like for example there's a substitution reaction.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000174 · EV0000175 · EV0000176 · +1 source units in pack.json

Core explanation · 05:00–09:59

Whereas if there was an elimination reaction, definitely some small molecules are getting eliminated and or even larger molecules sometimes and then what happens to your atom economy?

Key details

- It has become low.
- If you have an addition reaction, we know that.
- Talking of green synthesis, then of course we understand atom economy must be maximum, and that is 100%.
- Safer synthesis is the third principle.
- As far as possible, we should use only substances which are innocuous and necessary safety precautions have to be taken while processing to protect human health and the environment.
- But to prepare the isocyanate we had to use phosgene and we know that phosgene is a very toxic gas, COCl2.
- It's a toxic gas.
- So the alternative now has been found and the green synthesis of polyurethane eliminates the use of first gene and instead CO2 is being used.
- It functions as efficiency and as efficiently, but the toxicity is reduced.
- or in pharmaceuticals this plays a very important role because many of these drugs which are being used for as medicine we know that there is lot of side effects.
- Then design benign chemicals.
- And if it's.

- All we have to use, then they should be harmless, innocuous chemicals which are harmless.
- We can use alternative solvents, find alternative solvents greener in nature.
- So many of the researchers are using these ionic liquids now organic solvents are usually avoided because they are volatile in nature and they can affect the health as well as the environment but when you have ionic liquids now these have the properties to dissolve.
- At the same time they will not be volatile.
- We have pyridine which is like this.
- with three bonds and there is a plus because we have attach an alkyl group here.

Example or application from the lecture

- Suppose there's a double bond and the reactants are getting incorporated into it, yes then we get hundred percent atom economy, and even if there is a rearrangement reaction again it is just rearrangement of the atoms which is taking place again we have a hundred percent atom economy.
- For example, polyurethane is a very popular polymer and it was manufactured conventionally using isocyanate and alcohol.
- One good example would be N alkyl peridinium cation with BF4 minus.

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EV0000178 · EV0000179 · EV0000180 · +1 source units in pack.json

Core explanation · 10:00–14:59

This alkyl group could be ethyl, methyl, and so on.

Key details

- Positive is now because of the N having an extra bond over here, this becomes positive and this is neutralized with the B at four minus.
- At any cost we have to avoid harmful solvents like chloroform and peritene, etc., which are known to cause health hazards, and some of them could be carcinogenic to human beings and animals and also cause severe damage to the environment.
- The next principle says design for energy efficiency.
- Say the reaction could be occurring at milder conditions or it requires less time to complete.
- For this purpose, it is worthwhile to utili something like biocatalyst or homogeneous or heterogeneous catalyst, which will reduce the energy of activation for the reactions and hence we will consume less energy.
- And modern techniques such as solvent free super.
- And more importantly these days microwave irradium and ultrasound.
- Biodiesel when it is made in the lab, the normal way, it takes nothing less than say one hour one and a half hours.
- But the same thing when we do in the microwave when we use microwave radiation to have that reaction or to perform the reaction when then that reaction takes place at a very fast rate and.
- And within 15 minutes we can have the same reaction happening which was taking one and a half hours.

- less energy intensive and saving more energy.
- Use of renewable feedstock.
- I think all of us are aware of this.
- Raw material or feedstock used as starting material for any reaction should be a renewable so it's should come from a renewable source rather than one which is depleting.
- Wherever it is technically as well as economically possible and for renewable feedstock the most common thing being looked at is biomass and agricultural waste which are strongly recommended rather than using the non-reduable sources like coal and petroleum based raw materials which we know are depleting very fast.
- This is very important because many times when reactions are carried out, we make derivatives.
- So temporary modification should be avoided whenever possible because what happens because of that?
- Going to generate waste, and at the same time, we know whenever we have these extra byproducts getting form the atom efficiency is reduced as the protective group is not getting.
- Incorporated in the final product.
- like this and I want to nitrate it.
- The moment I do nitration I will get ortho as well as the para substituted but what I want is just the ortho or the para.
- I usually take aniline and I'm going to treat it with CH3CO C L, so thereby making it into NH.
- So because of this, what happened now?
- We are blocking, we are protecting the NH2 group or making it less reactive.
- And now when we do nitration, what we observe is that either we get the para.

Example or application from the lecture

- For example, biodiesel.
- For example, we could be using some group to block or as a protection and then again deprotection.
- Let us take an example.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000182 · EV0000183 · EV0000184 · +1 source units in pack.json

Core explanation · 15:00–19:59

Para product that is what we wanted.

Key details

- But then we wanted an ananine group, I mean NH2 group over here.
- in the paraposition and then NH2 nitroaniline that is para nitroaniline and ortho nitro aline aniline and not the disubstituted aniline.
- But what happened because of this you carried out so many reactions first of all and then you blocked this position and finally you have to unblock it like protection and deprotection

now because of that a lot of wastage happens so we have to look for alternatives wherein directly aniline can be converted into orthodonitromine or paranitry.
- Anuline as we desire.
- of chemicals and also the time consumed in the reaction.
- Now another way of doing carrying out reactions is by catalysis and this is definitely a greener method.
- So we have to use catalytic reactions because catalytic.
- This we know are going to speed up the reaction, they can be recycled and are highly selective in nature.
- we can get very high yield and also not only high yield but we'll get very pure products.
- So therefore wherever it is possible we should look for reactions which can be carried out with the help of a catalyst.
- B of greener options because when we use tychometric reactions, they are highly energy intensive and they also require excessive reagents.
- The next principles talk about design for degradation.
- We all understand that insecticides and pesticides they will remain in and they are required for the plants but nevertheless they remain in the soil and gets absorbed into the earth and from the water there when it comes into the water bod and finally it is going to come back to us and affect us in a bad way.
- They're going to cause harm.
- So therefore, insecticides, pesticides, polymers, etc.
- We have to look for degradable alternatives for this so that they do not persist in the environment and are known to cause these generally will persist in the environment and they are known to cause bioaccumulations.
- Therefore, we have to find alternatives which are degradable so that they do not persist in the environment.
- Then pollution prevention.
- Here we are talking of.
- Real time analysis.
- Of course, all these points which we have mentioned so far about are about pollution prevention, but here we are talking of real-time analysis that is quality control.
- The definition of hazardous substances.
- This is the last principle, that is the twelfth principle.
- The chemicals chosen for the processes should be such that.

Example or application from the lecture

- This is an example where we talk about we have to reduce derivatives so that we can bring down the wastage.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000186 · EV0000187 · EV0000188 · +1 source units in pack.json

Core explanation · 20:00–24:59

We can minimize accidents in the form of fires, explosions, and toxic releases to the environment.

Key details

- Also, manufacturing plants should be well equipped with safety mechanisms.
- All of us are aware of the mobile gas tragedy.
- It is one of the worst industrial disasters and after so many years people are still suffering because of the leakage of a toxic gas into the environment.
- It is a set of principles which emphasizes on environmentally benign chemical synthesis with a view to devise pathways for the prevention of pollution.
- and then bringing a paradigm shift in chemical processing to achieve sustainable development.
- It is a concept which we should always keep in mind whichever field we are working in, that we have to save the environment, we have to keep it clean and safe.
- That brings us to the end of module eight and we have completed unit five.
- It also brings us to the end of this course on engineering chemistry.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000190 · EV0000191 · EV0000192 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Green chemistry: Introduction, twelve principles of green chemistry with examples transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__65cb11f75f072aae.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Green chemistry: Introduction, twelve principles of green chemistry with examples transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__65cb11f75f072aae.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Green chemistry: Introduction, twelve principles of green chemistry with examples transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__65cb11f75f072aae.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9983; only the JSON segmentation supplies the displayed timestamps.
Green chemistry: Introduction, twelve principles of green chemistry with examples transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__65cb11f75f072aae.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00046	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...
SRC00047	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00048	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...
SRC00049	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...

C Revision and retrieval practice